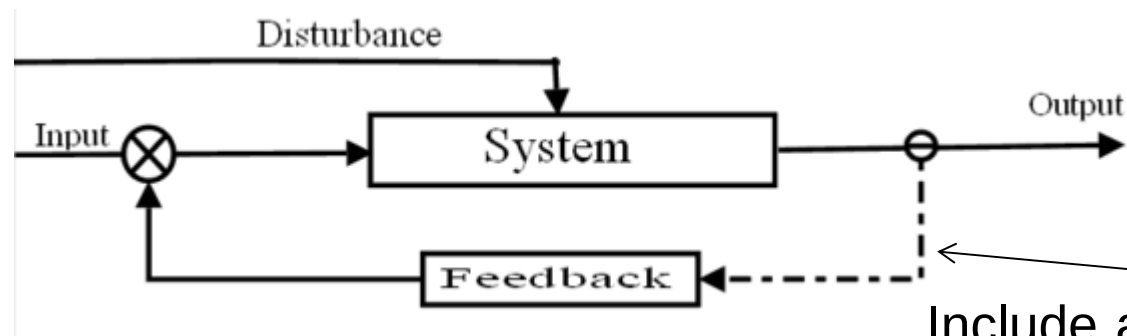
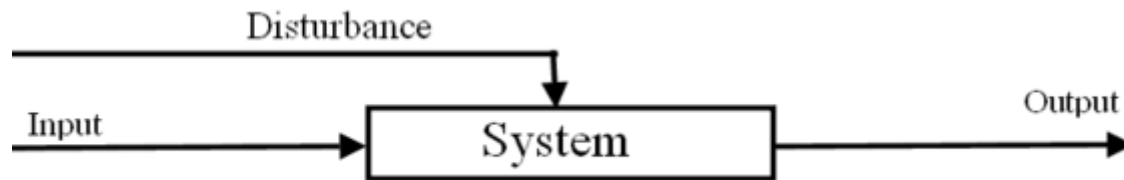


Building a Primitive pilot model *proportional-integral-derivative* *(PID)* in the 3-dof helicopter model

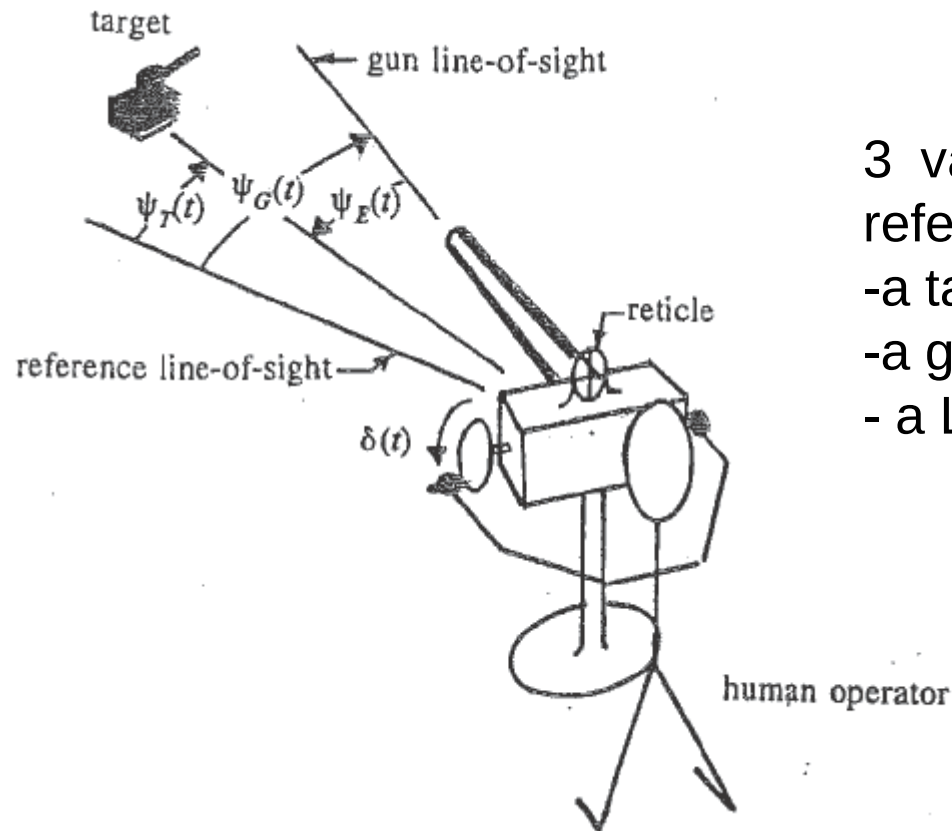
Open loop and closed loop control



Include a negative feedback (feedback loop) to control the dynamic behaviour of the system

Example of feedback control: Gunnery azimuth tracking task (1947)

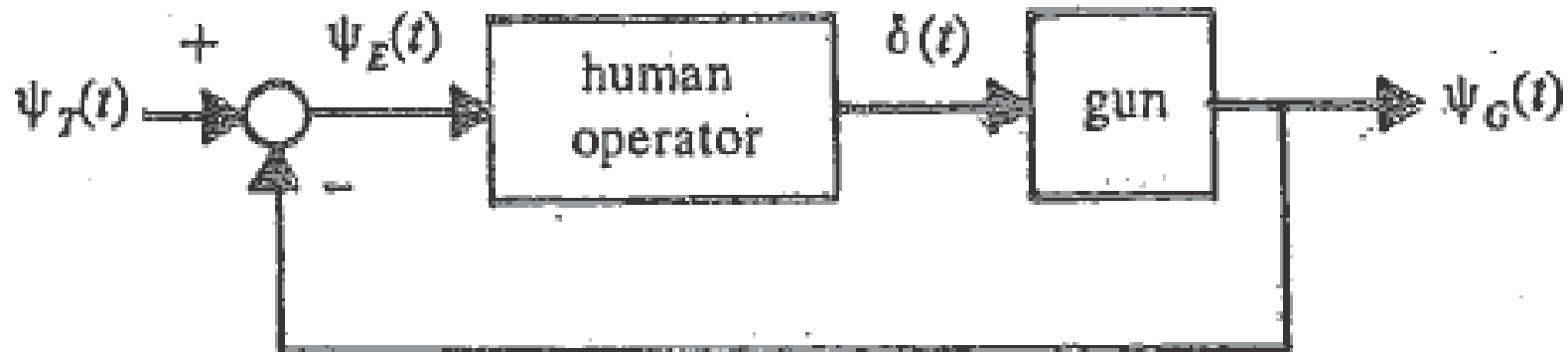
Tustin (1947) pioneering study of the human-in-the-loop control



3 variables can be defined from the reference line-of-sight (LOS):

- a target LOS $\Psi_T(t)$
- a gun LOS $\Psi_G(t)$
- a LOS error $\Psi_E(t) = \Psi_T(t) - \Psi_G(t)$

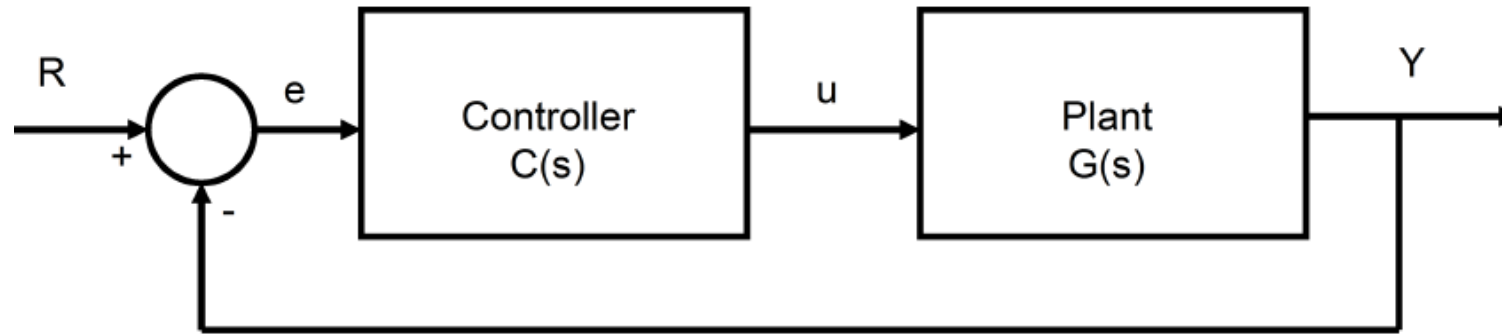
Gunnery azimuth tracking task



Why learn PID controller?

- Because PID Controllers are everywhere
- Due to their simplicity they can be tuned for optimal performance in many applications
- PID controllers are used in more than 95% of closed-loop industrial processes.
- They can be tuned by operators without extensive background in controls

PID-controller structure



A PID controller attempts to correct an "error" e

$$e(t) = R - Y$$

R = reference value (desired value, (setpoint))

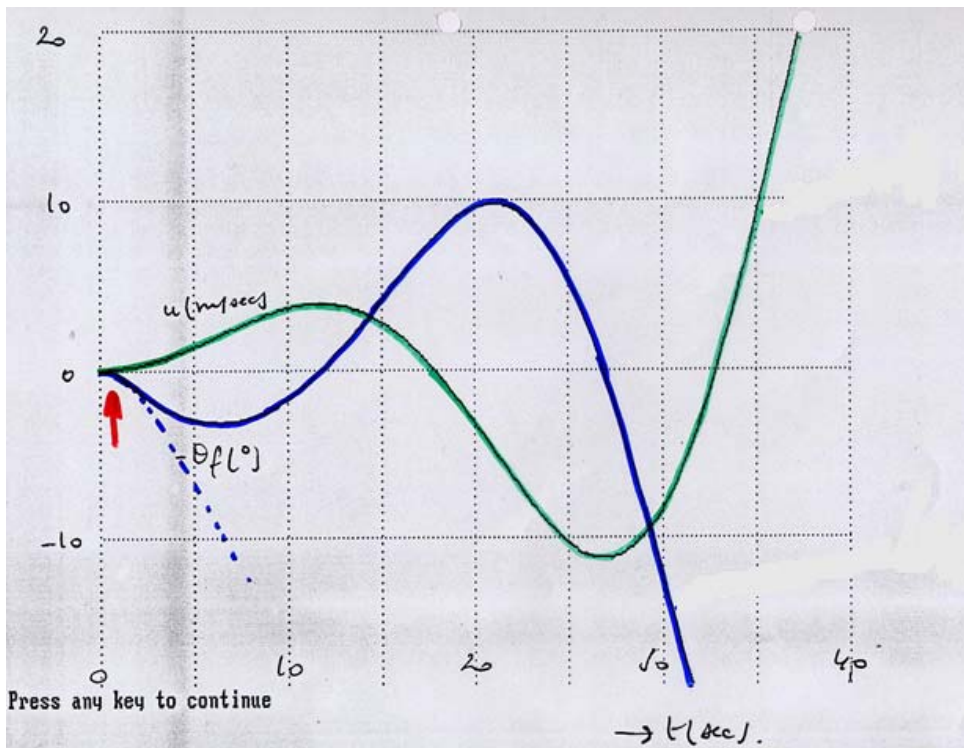
Y = System output (sensed value, current value)

$u(t)$ = system input

“Keeping” the hover in the 3-dof model

Helicopter Open loop flight

Disturbance from hover $0.5 < t < 1$ sec with $q_c = 1$ deg



$$\dot{u} = -g \sin \theta_f - \frac{D_f}{m} \cdot \frac{u}{V} + \frac{T}{m} \sin(\theta_c - a_1) - qw$$

$$\dot{w} = g \cos \theta_f - \frac{D_f}{m} \cdot \frac{w}{V} - \frac{T}{m} \cos(\theta_c - a_1) + qu$$

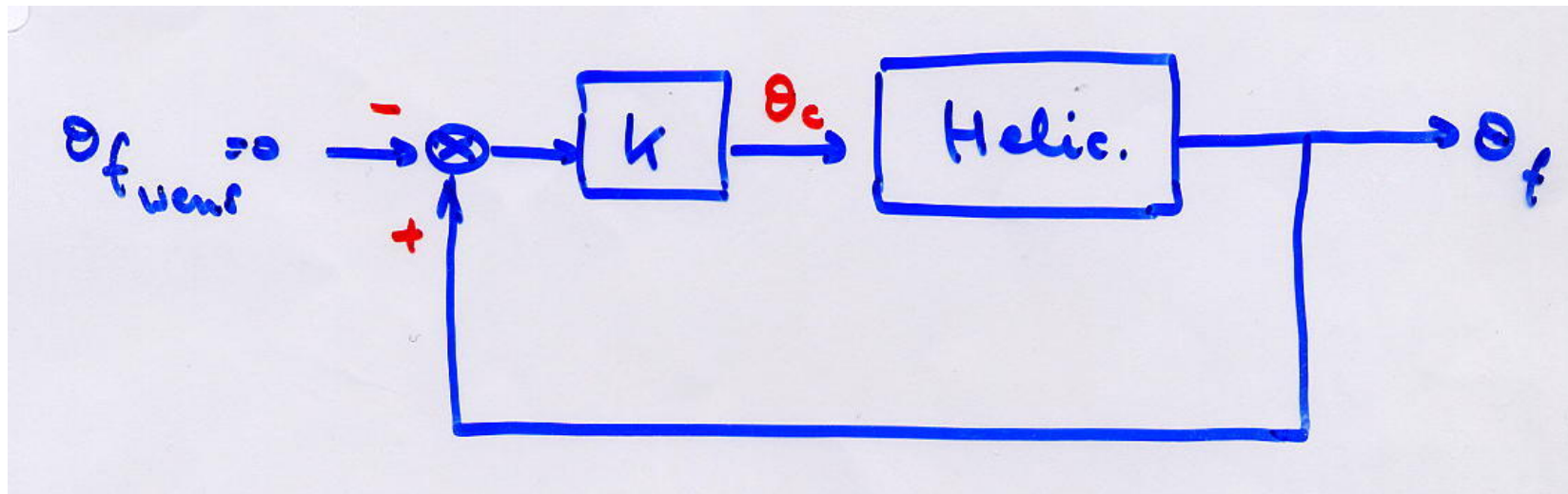
$$\dot{q} = -\frac{T}{I_y} h \sin(\theta_c - a_1)$$

$$\dot{\theta}_f = q$$

P-controller

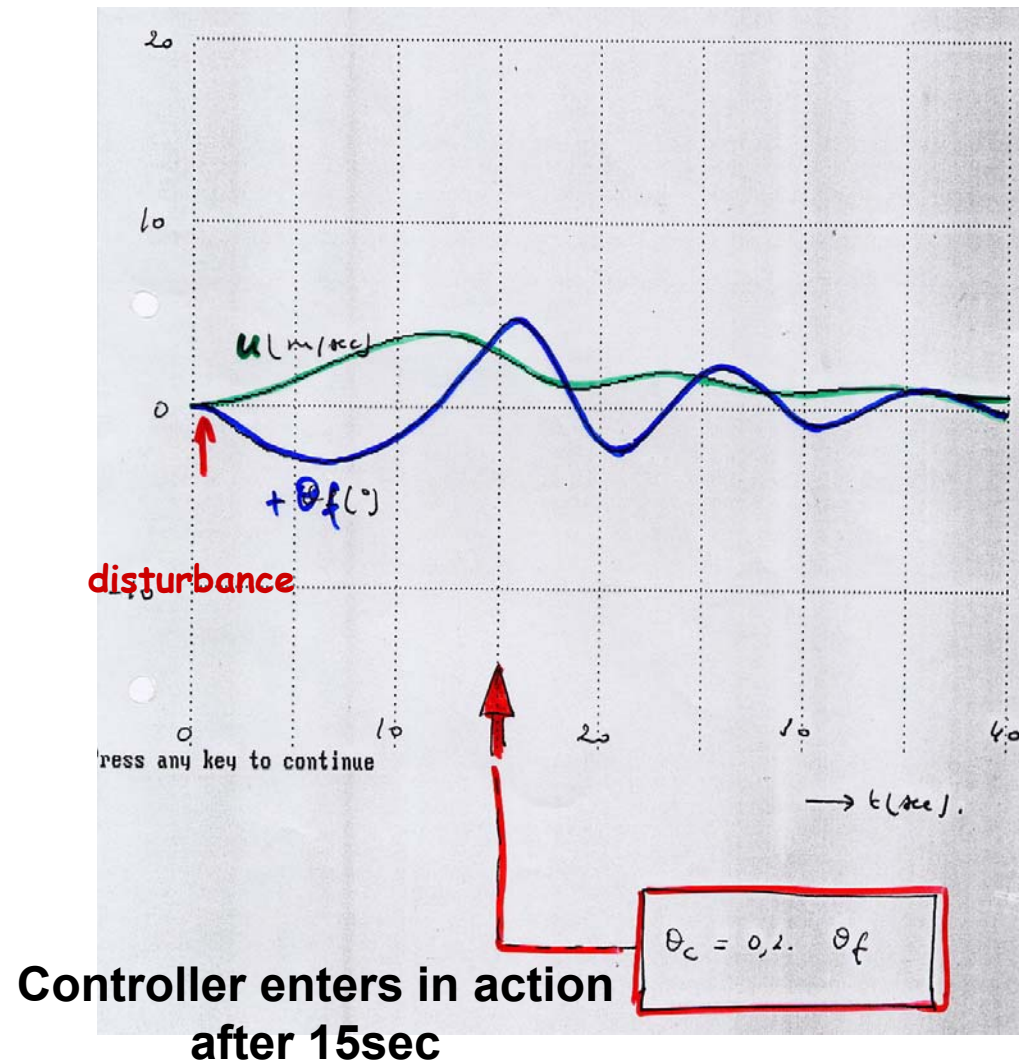
P-controller for cyclic pitch

Hovering flight

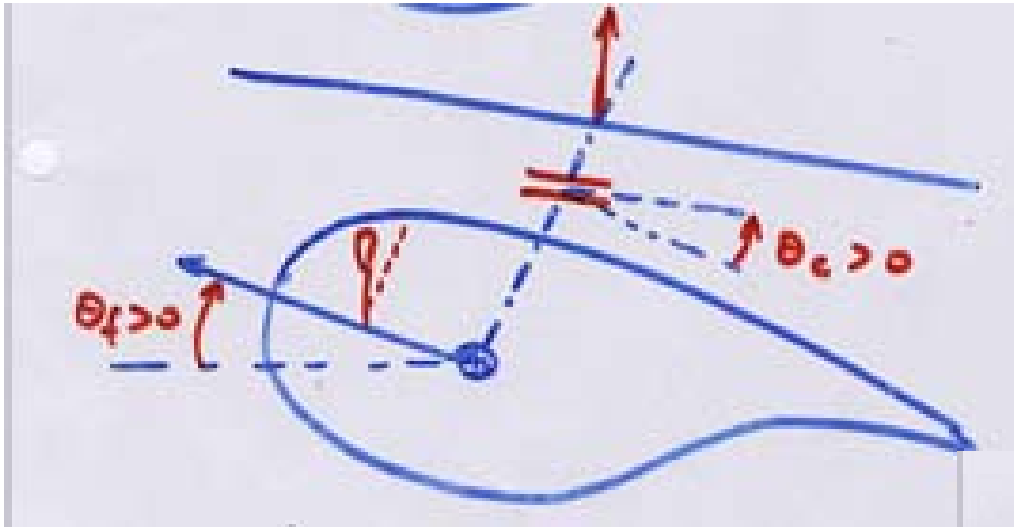


$$\theta_c = K \cdot (\theta_f - \theta_{f\text{ wens}})$$

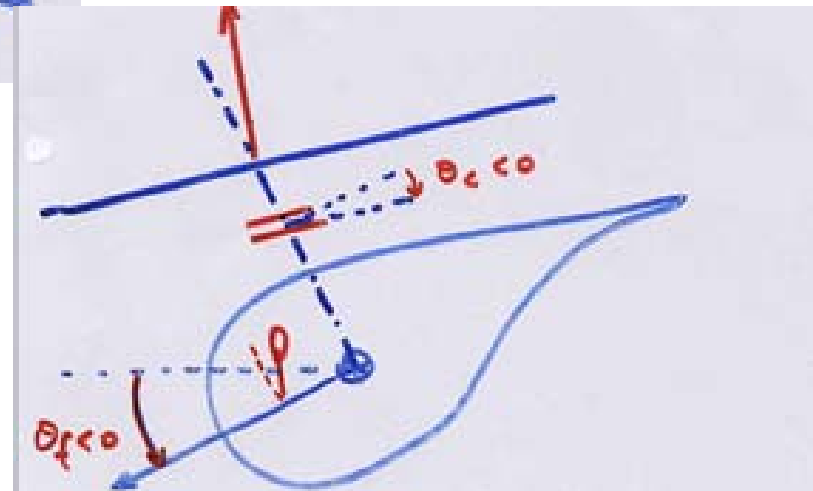
P-controller for cyclic pitch



P-controller for cyclic pitch

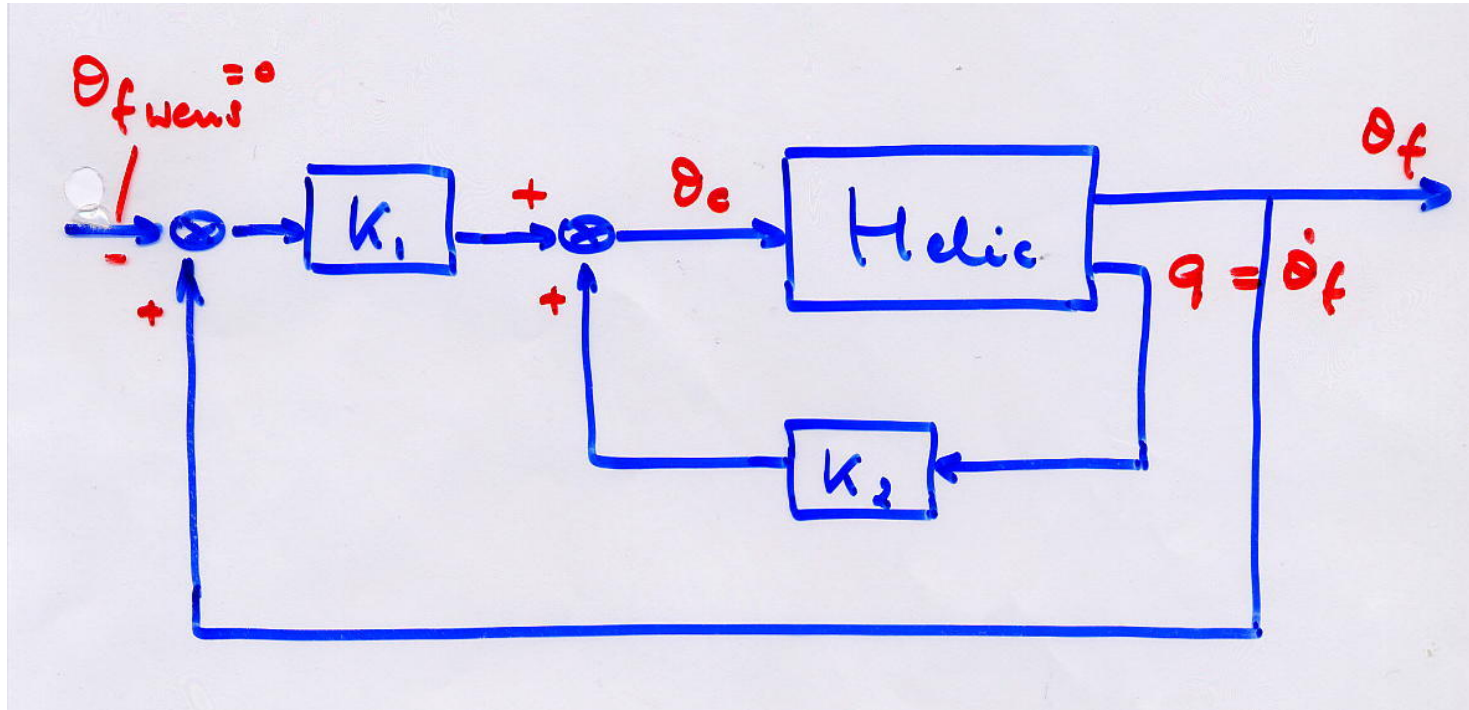


Static stability, but possible dynamic instability



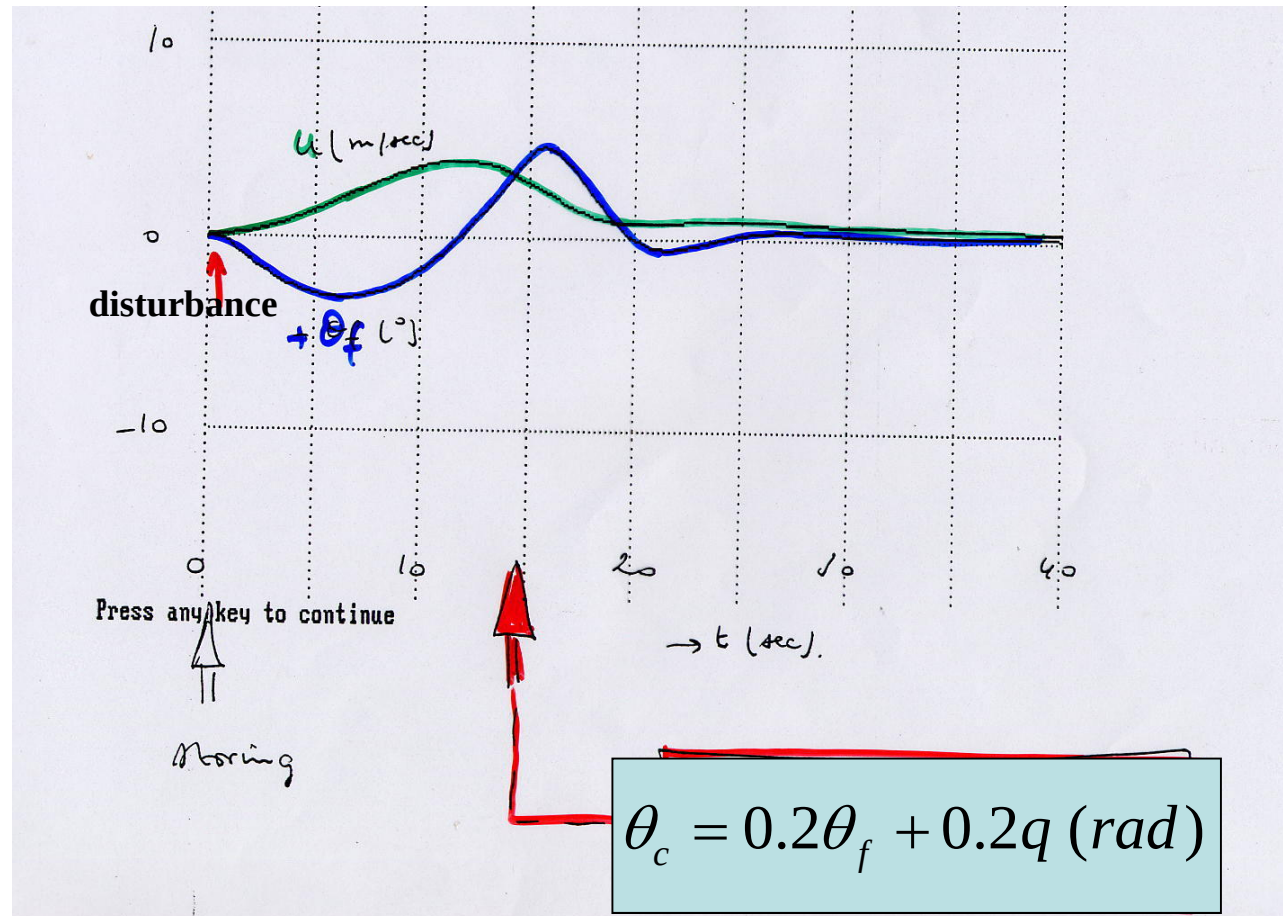
D-controller

PD-controller for cyclic pitch

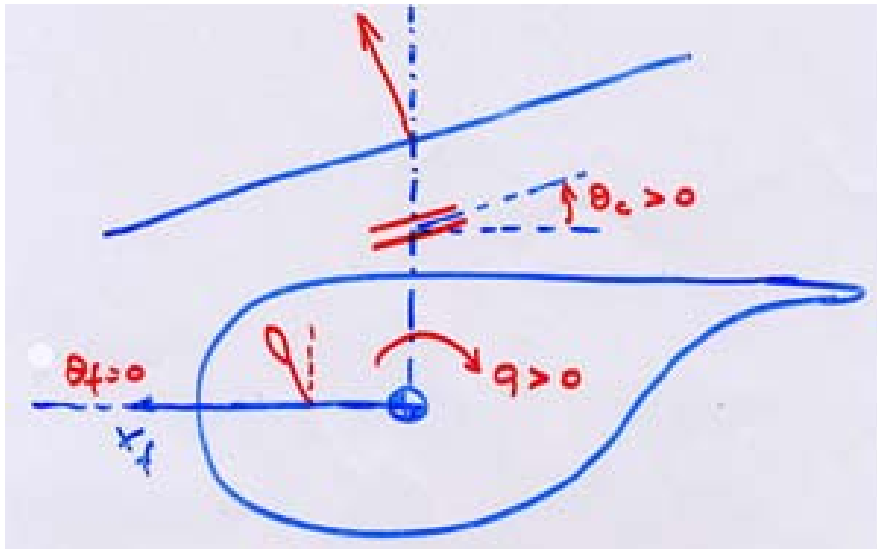


$$\theta_c = K_1 \cdot (\theta_f - \theta_{f \text{ wens}}) + K_2 \cdot q$$

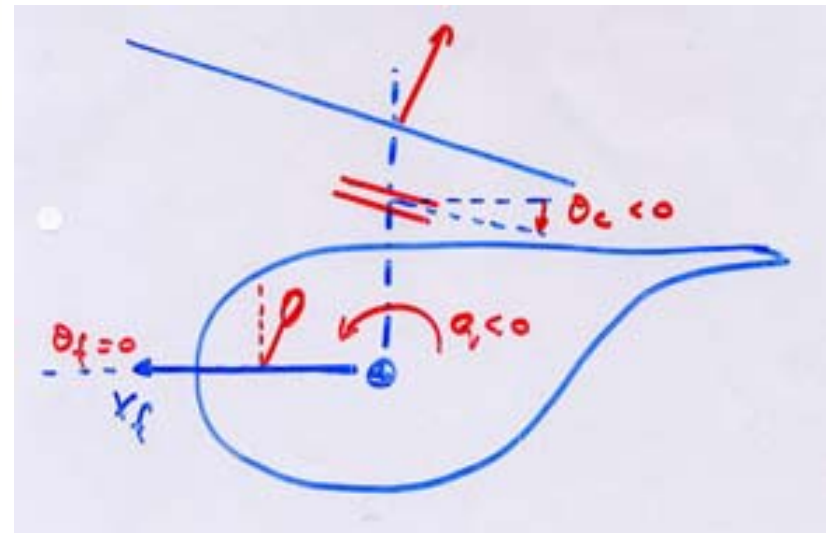
PD-controller for cyclic pitch



PD-controller for cyclic pitch

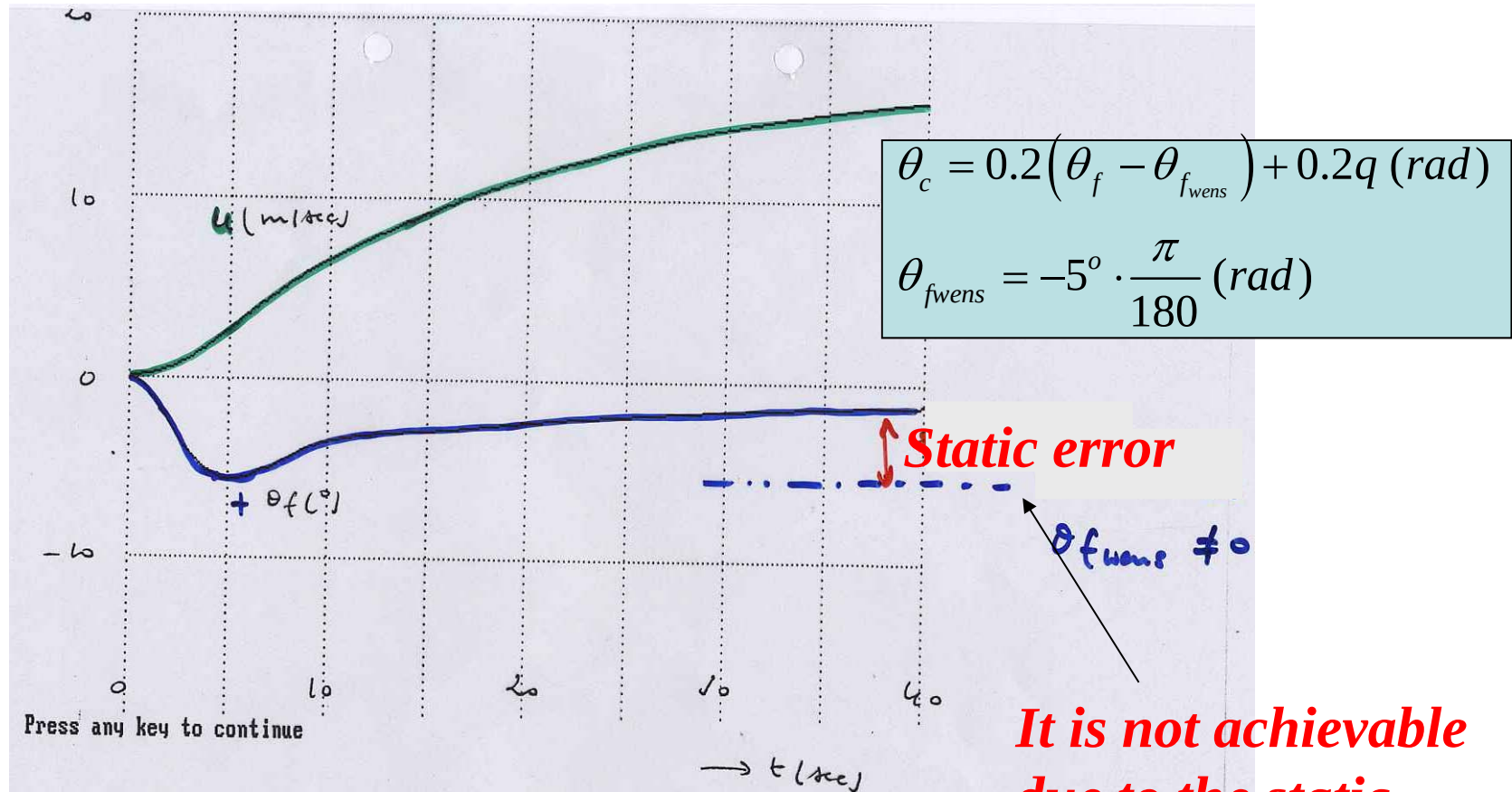


Improved pilot model but still a problem: $u > 0$ and we have drifted away from the hovering point



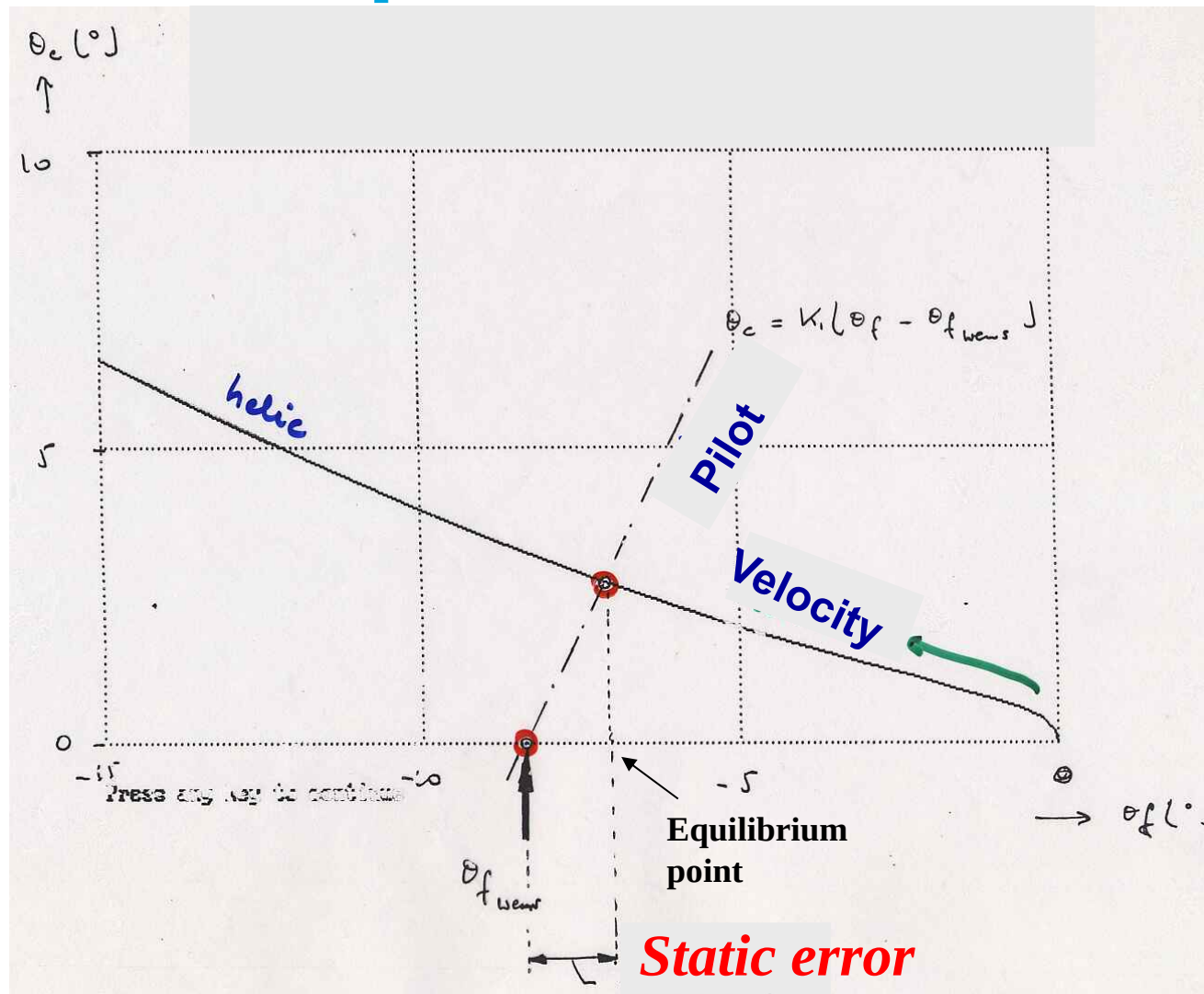
**Going 5 deg nose
down desired attitude
from hover**

PD controller for achieving 5 deg nose down desired attitude



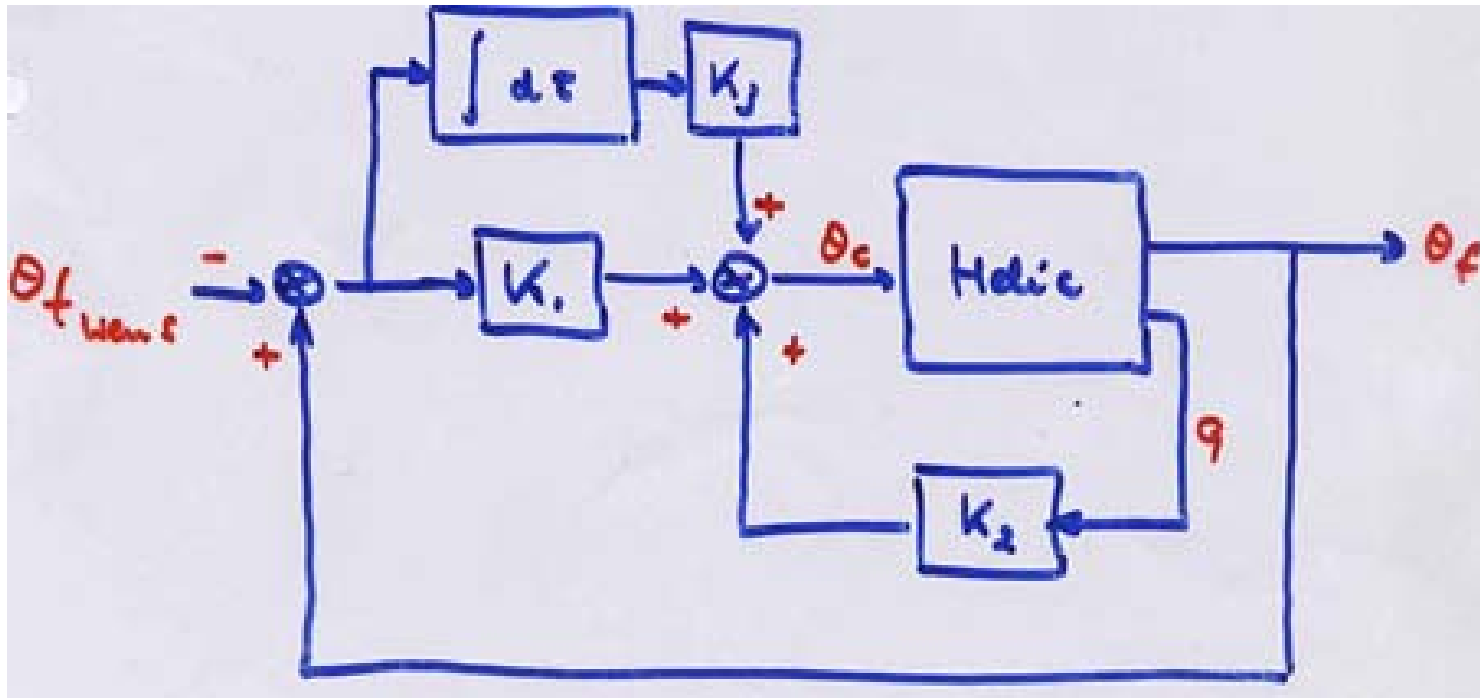
*It is not achievable
due to the static
error of the control*

How to explain the static error?



I-controller

PID controller for cyclic pitch

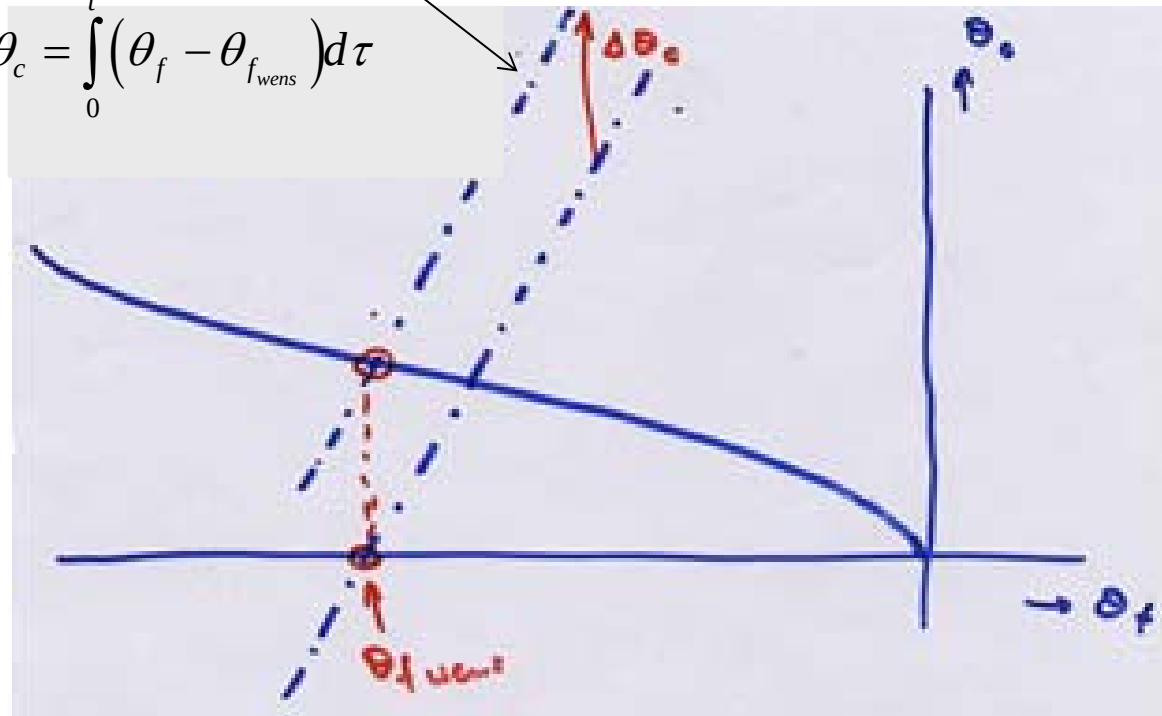


$$\theta_c = K_1 \left(\theta_f - \theta_{fwens} \right) + K_2 q + K_3 \int_0^t \left(\theta_f - \theta_{fwens} \right) d\tau \text{ (rad)}$$

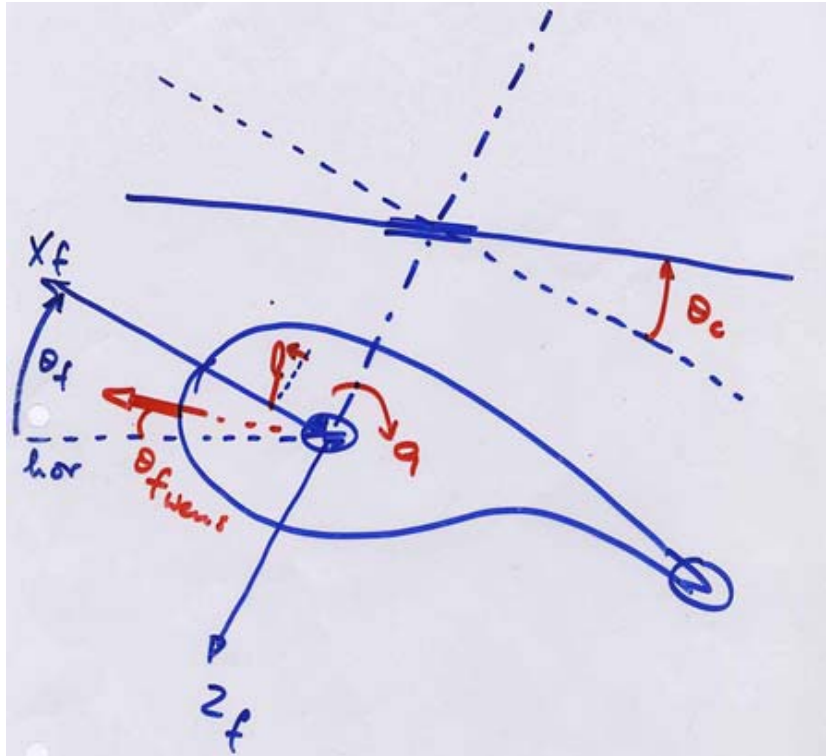
PID controller for cyclic pitch

$$\theta_c = K(\theta_f - \theta_{fwens}) + \Delta\theta_c \text{ (rad)}$$

$$\Delta\theta_c = \int_0^t (\theta_f - \theta_{fwens}) d\tau$$



PID controller for cyclic pitch

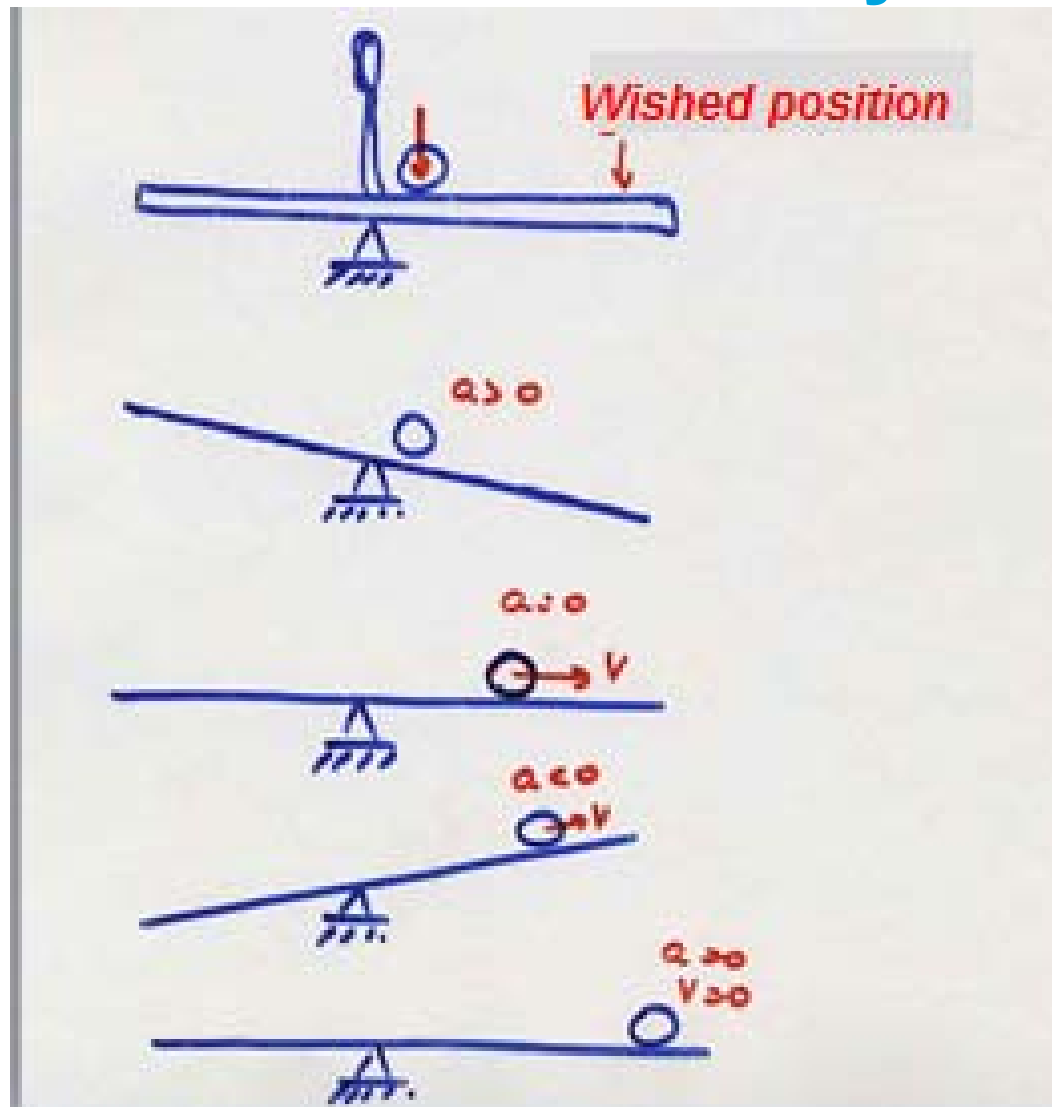


A PID controller requires 2 integration levels for humans. This is done unconsciously by the pilot spinal cord and required training

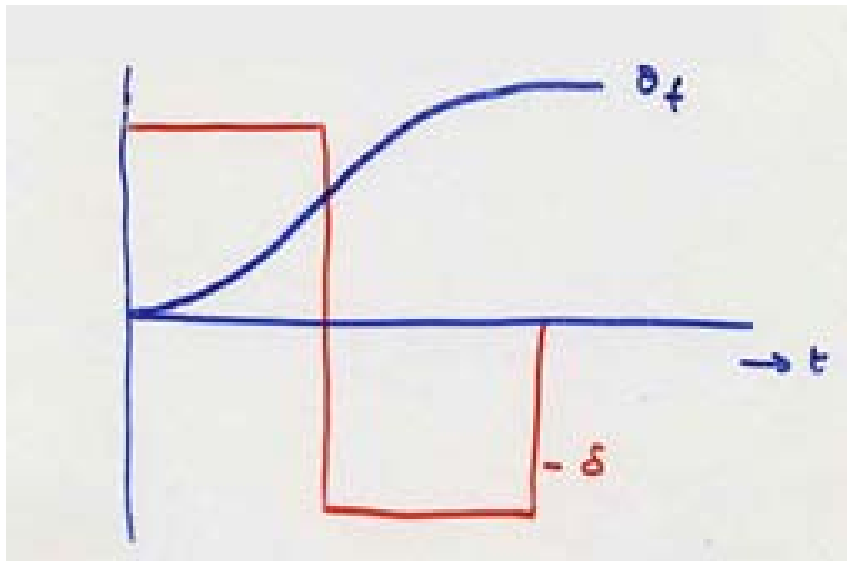
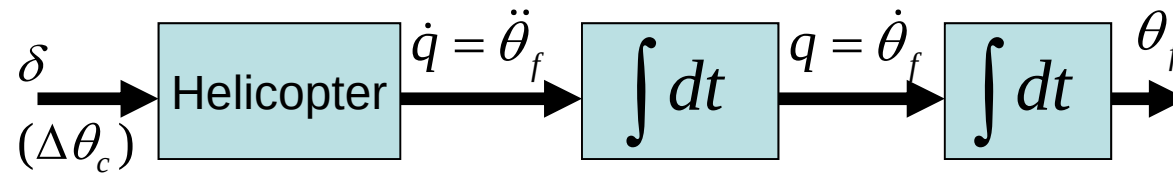
The conscious part of the manoeuvre (pitching the helicopter to a θ_{wens}) is done by the pilot brain

$$\theta_c = K_1 (\theta_f - \theta_{fwens}) + K_2 q + K_3 \int_0^t (\theta_f - \theta_{fwens}) d\tau$$

Acceleration-control system

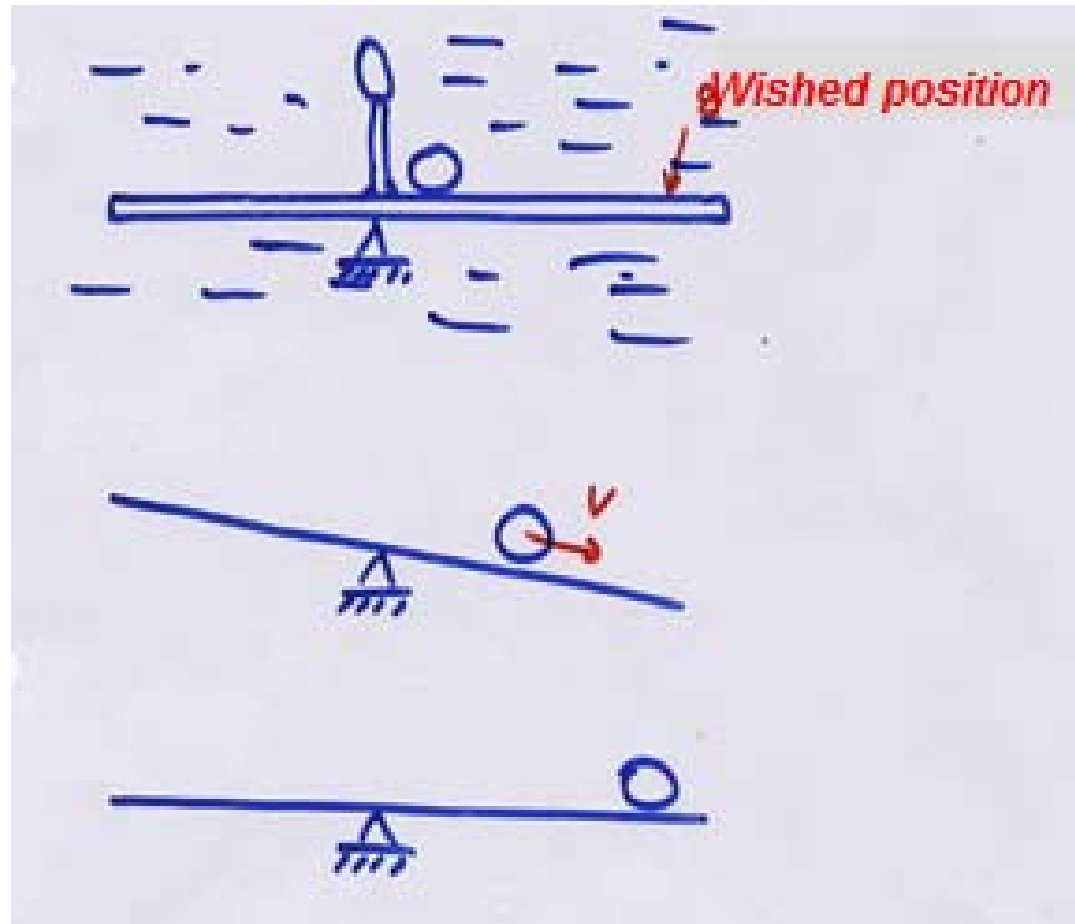


Acceleration-control system

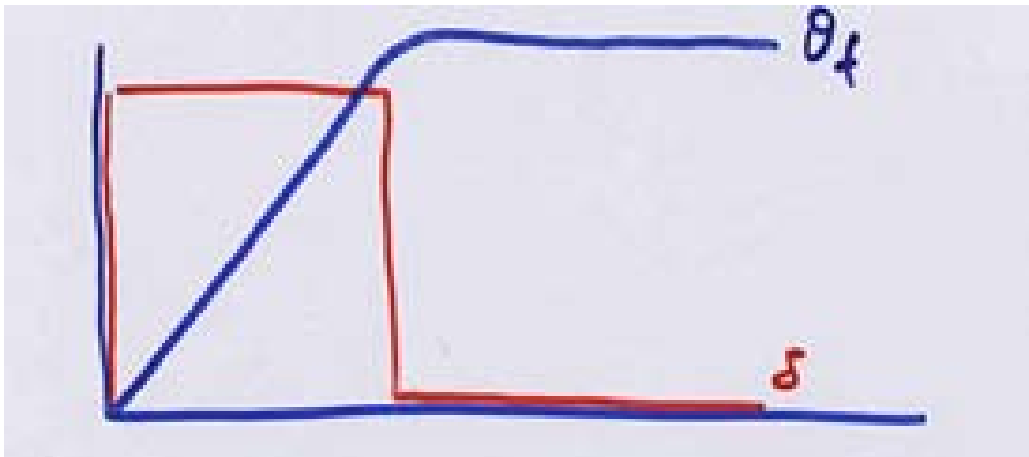
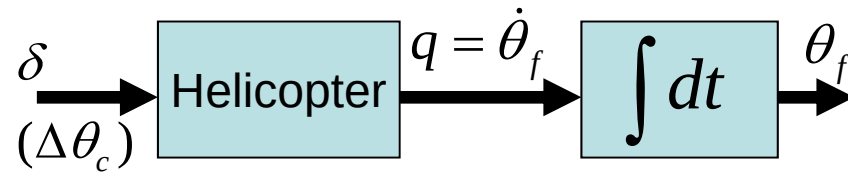


Command to the wished fuselage attitude needs 2 integration levels for anticipation

Rate-control system

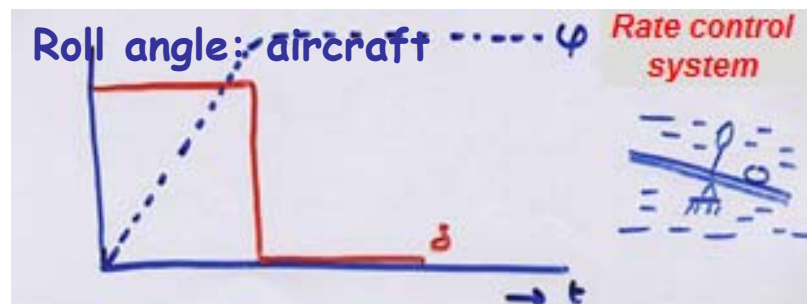
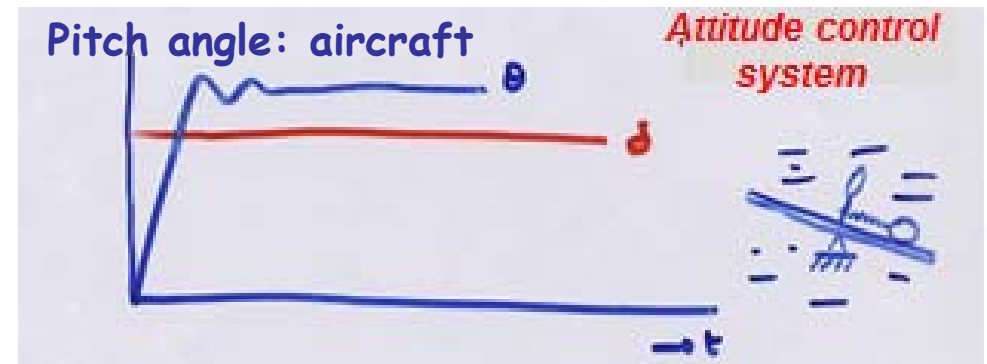
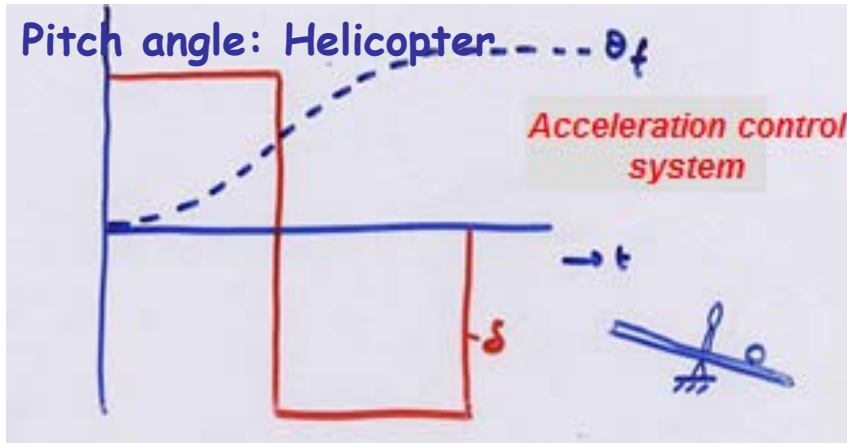


Rate-control system

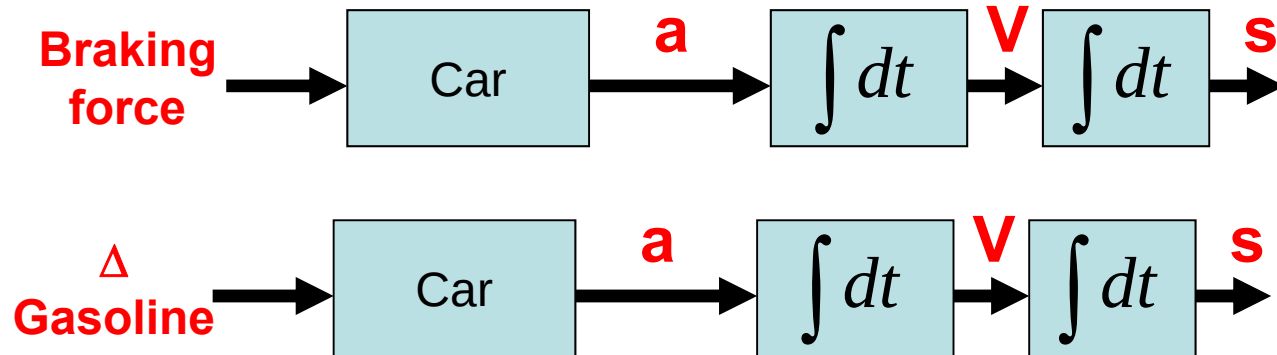


Command to the wished fuselage attitude needs one integration level for anticipation

Comparison between helicopter and fixed wing control



Comparison to the case of a car



⇒ 1st Integration level: Anticipation on the free road

⇒ 2nd Integration level: Anticipation in the traffic jam

⇒ **Helicopter pitch/roll control in forward flight**
≡ Car in the traffic jam